

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	
Project Title	Rising Waters: Smart AI Flood Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to develop an **AI-powered Flood Prediction System** using machine learning algorithms to provide early flood warnings based on historical weather and rainfall data. The system analyzes meteorological parameters such as annual rainfall, seasonal rainfall, cloud visibility, and other environmental factors to predict flood risk accurately. By delivering timely predictions, the application helps disaster management authorities make informed decisions, minimize property damage, and save lives. The solution integrates multiple machine learning models and deploys the best-performing model through a Flask web application for real-time flood risk assessment

Project Overview	
Objective	To develop an intelligent flood prediction system using machine learning techniques that accurately predicts flood occurrences and provides early warning alerts for disaster preparedness.
Scope	The project focuses on analyzing historical weather data, training multiple machine learning models, selecting the most accurate model (XGBoost), and deploying it through a Flask web application for real-time flood prediction.
Problem Statement	
Description	Existing flood forecasting systems often lack sufficient accuracy and timely predictions, resulting in delayed emergency response, increased property damage, and loss of human life during flood events.
Impact	Accurate flood prediction enables early evacuation, efficient disaster management, optimized resource allocation, reduced economic losses, and improved public safety.
Proposed Solution	
Approach	Develop a machine learning-based flood prediction system using Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and XGBoost algorithms. Train the models using historical weather datasets, evaluate their performance, and deploy the best-performing model within a Flask web application for real-time flood risk prediction.

Key Features	• AI-based flood prediction using weather data. • Early warning system for flood-prone areas. • Multiple machine learning model comparison. • XGBoost model with 96.55% accuracy. • User-friendly Flask web interface. • Real-time flood risk prediction. • Scalable deployment on IBM Cloud. • Supports disaster management and resource planning.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 (or equivalent), 4–8 CPU Cores, Optional NVIDIA GPU (T4/GTX 1650 or higher)
Memory	RAM specifications	8 GB RAM (Minimum), 16 GB Recommended
Storage	Disk space for data, models, and logs	20 GB Available Storage, 512 GB SSD or Higher
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, XGBoost, pandas, numpy, matplotlib, joblib
Development Environment	IDE	Visual Studio Code / PyCharm / Jupyter Notebook
Data		
Data	Source, size, format	Historical Flood Prediction Dataset (Kaggle), ~6,000+ records, CSV Format

